

Тригонометрические многочлены и экстремальные задачи

Домашнее задание

1 сентября 2026 года

Теоретический минимум. Trigonometric polynomials and their zeros; the Dirichlet and Fejér kernels; Fejér's theorem and the Weierstrass approximation theorem; Fejér–Riesz factorisation of non-negative polynomials; Bernstein's inequality $\|T'\| \leq n\|T\|$; Parseval's identity; Lebesgue constants.

1. Prove Bernstein's inequality $\|T'\|_\infty \leq n\|T\|_\infty$ for trigonometric polynomials of degree n , and show that equality holds exactly for $T = c \sin(n\theta + \varphi)$.
2. (Fejér) Prove that the maximum of a_1 over all non-negative cosine polynomials $1 + \sum_{k=1}^n a_k \cos k\theta$ equals $2 \cos \frac{\pi}{n+2}$, and find the extremal polynomial.
3. (MIPT 2018) Let m and n be odd positive integers. Determine the sign of $I(y) = \int_0^\pi \sin^n x \cdot \sin^m(x+y) dx$ as a function of the parameter y .
4. (RMO) Let M be the set of continuous $f: [0, \pi] \rightarrow \mathbb{R}$ with $\int_0^\pi f(x) \sin x dx = \int_0^\pi f(x) \cos x dx = 1$. Find $\min_{f \in M} \int_0^\pi f^2(x) dx$.
5. (неравенство Никольского) Prove that $\|T\|_\infty \leq \sqrt{\frac{2n+1}{2\pi}} \|T\|_{L^2[0, 2\pi]}$ for every trigonometric polynomial of degree n , and show that equality is attained exactly on the shifted Dirichlet kernels.
6. Prove Parseval's identity for trigonometric polynomials, then compute $\sum_{k \geq 1} k^{-2}$ and $\sum_{k \geq 1} k^{-4}$ from the Fourier series of x and x^2 on $[-\pi, \pi]$.
7. Prove that the Lebesgue constants $L_n = \frac{1}{2\pi} \int_{-\pi}^\pi |D_n(\theta)| d\theta$ of the Dirichlet kernel satisfy $L_n = \frac{4}{\pi^2} \ln n + O(1)$, and deduce that there is a continuous function whose Fourier series diverges at a point.

8. Prove that a non-negative trigonometric polynomial vanishing at θ_0 has a zero of even multiplicity there, and deduce that a non-negative polynomial of degree n has at most n distinct zeros.
9. (Turán) Prove that if T is a trigonometric polynomial of degree n with $\|T\|_\infty = 1$, then T takes values of modulus at least $\frac{1}{2}$ on a set of measure at least c/n ; find an explicit constant c .
10. Prove the Marcinkiewicz–Zygmund discretisation: for a trigonometric polynomial of degree n

$$\frac{1}{2n+1} \sum_{j=0}^{2n} \left| T\left(\frac{2\pi j}{2n+1}\right) \right|^2 = \frac{1}{2\pi} \int_0^{2\pi} |T(\theta)|^2 d\theta,$$

and explain why the same identity fails for $2n$ nodes.